DD_FUL-10 - Technology Migration Fulfillment - Developer Implementation Guide

Version: v1.0 **Bundle:** 06_fulfillment **Companion DD:** DD_FUL-10-Technology_Migration_Fulfillment-v1.0

1. Goal

Implement the technical fulfillment flow for access technology migration.

The important separation is:

- SUB-WF-MIGRATION starts and commits the subscription operation.
- FUL-10 validates and executes the technical migration.
- WO-01 executes technician work.
- OSR/OSR-INSTANCE own stock and serial state.
- FUL-09 handles equipment swap/rebind fulfillment when needed.
- FUL-03/PROV-INT send network commands.

2. Runtime Flow

```
sequenceDiagram
  participant SUB as SUB-WF-MIGRATION
  participant FUL10 as FUL-10
  participant RLM as RLM-CFG/HomePass
  participant OSR as OSR/OSR-INSTANCE
  participant WO as WO-01
  participant FUL09 as FUL-09
  participant PROV as FUL-03/PROV-INT
  participant APR as EM-CFG-04

  SUB->>FUL10: POST /api/technology-migrations/fulfillments
  FUL10->>RLM: read source/target HomePass capability
  FUL10->>OSR: read expected/available equipment
  FUL10->>FUL10: run Drools eligibility/equipment policy
  FUL10->>WO: POST /api/work-orders kind TECHNOLOGY_MIGRATION
  WO-->>FUL10: WorkOrderFinalized with field evidence
  FUL10->>FUL09: start equipment swap fulfillment when required
  FUL10->>PROV: activate target technology
  FUL10->>PROV: suspend/teardown source technology
  FUL10->>PROV: reconcile
  FUL10-->>SUB: fulfillment outcome
  FUL10->>APR: approval only for override/force-complete/rollback
```

3. Step 1 - Database Migrations

Create:

- `technology_migration_fulfillment`
- `technology_migration_validation_check`
- `technology_migration_equipment_plan`
- `technology_migration_cutover_step`
- `technology_migration_exception`

Required constraints:

- `technology_migration_fulfillment(operator_code, subscription_operation_id)` unique.
- One active technology migration per `(operator_code, subscription_id)`.
- Foreign references may be soft references because modules are separate, but store enough ids for support and reconciliation.

Add table and column comments in the migration because support teams will inspect these tables during failed cutovers.

4. Step 2 - Receive Request From SUB-WF

Endpoint:

```
POST /api/technology-migrations/fulfillments
Idempotency-Key: tmf-op-mig-001
Content-Type: application/json
```

Request:

```
{
  "operatorCode": "WIK",
  "subscriptionOperationId": "op-mig-001",
  "subscriptionId": "SUB-700045",
  "accountId": "ACC-900045",
  "customerId": "CUS-100045",
  "sourceTechnology": "HFC",
  "targetTechnology": "GPON",
  "sourceHomepassId": "HP-HFC-0100",
  "targetHomepassId": "HP-GPON-0440",
  "targetPackageRef": "WIK-FIBER-50M",
  "migrationType": "ACCESS_TECH_CHANGE"
}
```

Implementation:

1. Validate required fields and enum values.
2. Check idempotency by `operator_code + subscription_operation_id`.
3. If row exists with the same request hash, return existing row.
4. If row exists with a different request hash, return `409`.
5. Check no other active FUL-10 fulfillment exists for the subscription.
6. Insert `technology_migration_fulfillment` with `REQUESTED`.
7. Emit `TechnologyMigrationFulfillmentRequested`.

Do not call procurement. Missing stock must be handled before or during WO/OSR reservation; OSR-02 is not a runtime customer migration dependency.

5. Step 3 - Validate Source And Target Context

Endpoint:

```
POST /api/technology-migrations/fulfillments/tmf-001/validate
```

FUL-10 reads:

- source and target HomePass from RLM-CFG-01;
- target package requirements from SIP/PLM catalog APIs already used by SUB-WF;
- current subscription summary from SUB-LM/SUB-WF read API if needed;
- expected equipment and current bindings from OSR-INSTANCE;
- service provisioning capability from FUL-03/PROV-INT metadata if available.

Create validation rows for:

- `SOURCE_HOMEPASS_ACTIVE`
- `TARGET_HOMEPASS_ACTIVE`
- `TARGET_TECH_SUPPORTS_PACKAGE`
- `TARGET_CAPACITY_AVAILABLE`
- `EQUIPMENT_PLAN_AVAILABLE`
- `PROVISIONING_ADAPTER_AVAILABLE`

- `BILLING_READY_FOR_MIGRATION`

Run Drools:

```
public record TechnologyMigrationFact(
    String operatorCode,
    String sourceTechnology,
    String targetTechnology,
    String migrationType,
    boolean targetSupportsAllRequiredServices,
    boolean targetHasCapacity,
    boolean equipmentAvailable,
    boolean customerVisitRequired,
    boolean sameBuilding,
    boolean plannedAreaCutover
) {}
```

```
package rules.fulfillment.technology_migration.eligibility

rule "Target must support required services"
when
  $f : TechnologyMigrationFact(targetSupportsAllRequiredServices == false)
  $d : TechnologyMigrationDecision()
then
  $d.reject("TARGET_TECH_PACKAGE_INCOMPATIBLE");
end

rule "GPON migration needs target capacity"
when
  $f : TechnologyMigrationFact(targetTechnology == "GPON", targetHasCapacity == false)
  $d : TechnologyMigrationDecision()
then
  $d.reject("TARGET_GPON_CAPACITY_NOT_AVAILABLE");
end
```

If all required checks pass, update fulfillment to `VALIDATED`.

6. Step 4 - Build Equipment Plan

FUL-10 should decide whether the migration needs equipment work.

Examples:

```
package rules.fulfillment.technology_migration.equipment

rule "HFC to GPON requires old modem recovery and new ONT install"
when
  $f : TechnologyMigrationFact(sourceTechnology == "HFC", targetTechnology == "GPON")
  $d : TechnologyMigrationDecision()
then
  $d.addEquipmentAction("SOURCE_RECOVERY", "RECOVER_ONLY");
  $d.addEquipmentAction("TARGET_INSTALL", "CALL_FUL_09");
end

rule "Same CPE can be reused only when technology and SKU capability match"
when
  $f : TechnologyMigrationFact(sourceTechnology == targetTechnology)
  $d : TechnologyMigrationDecision()
then
  $d.addEquipmentAction("REUSE", "NO_ACTION");
end
```

Implementation:

1. Read current equipment bindings from OSR-INSTANCE.
2. Read required target equipment profile from package/catalog.
3. Run equipment rules.
4. Insert `technology_migration_equipment_plan` rows.
5. If required equipment is not available/reserved, fail validation or request WO/OSR reservation.
6. If equipment swap fulfillment is needed, call FUL-09 after field readiness or when WO assigns the target serial.

FUL-10 does not mutate `stock_balance` or equipment instance state directly.

7. Step 5 - Create Work Order

Endpoint:

```
POST /api/technology-migrations/fulfillments/tmf-001/create-work-order
Content-Type: application/json
```

Request:


```
{
  "preferredScheduleFrom": "2026-08-04T08:00:00Z",
  "preferredScheduleTo": "2026-08-04T13:00:00Z",
  "requiresCustomerVisit": true
}
```


FUL-10 calls WO-01:

```
POST /api/work-orders
Idempotency-Key: tmf-001-wo
Content-Type: application/json
```


```
{
  "operatorCode": "WIK",
  "kind": "TECHNOLOGY_MIGRATION",
  "originatingContextType": "TECHNOLOGY_MIGRATION_FULFILLMENT",
  "originatingContextId": "tmf-001",
  "subscriptionId": "SUB-700045",
  "accountId": "ACC-900045",
  "homepassId": "HP-GPON-0440",
  "metadata": {
    "sourceTechnology": "HFC",
    "targetTechnology": "GPON",
    "targetPackageRef": "WIK-FIBER-50M",
    "requiredEvidence": ["targetSerial", "port", "photo", "gps"]
  }
}
```


Update FUL-10 status to `WO_CREATED`.

When WO finalizes, it must provide:

- target serial or equipment instance id;
- port/splitter/tap information;
- installation evidence;
- technician completion status;
- failure reason if not completed.

8. Step 6 - Execute Cutover

Endpoint:

```
POST /api/technology-migrations/fulfillments/tmf-001/execute-cutover
Content-Type: application/json
```

Request:


```
{
  "fieldCompletionRef": "wo-mig-001",
  "targetEquipmentInstanceId": "inst-gpon-001",
  "cutoverMode": "MAKE_BEFORE_BREAK"
}
```


Build cutover steps using Drools:

```
package rules.fulfillment.technology_migration.cutover

rule "HFC to GPON make-before-break"
when
  $f : TechnologyMigrationFact(sourceTechnology == "HFC",
                               targetTechnology == "GPON")
  $d : TechnologyMigrationDecision()
then
  $d.setCutoverMode("MAKE_BEFORE_BREAK");
  $d.addStep("GPON_ACTIVATE_TARGET_ONT");
  $d.addStep("GPON_RECONCILE_TARGET");
  $d.addStep("HFC_SUSPEND_SOURCE_MODEM");
  $d.addStep("HFC_RECONCILE_SOURCE");
```

```
end
```

For every step:

1. Insert `technology_migration_cutover_step` in `PENDING`.
2. Send command through PROV-INT/FUL-03.
3. Store request/response snapshots.
4. Update step status.
5. Stop on failure unless rule policy allows next step.

Sample provisioning command:

```
POST /api/provisioning/commands
Idempotency-Key: tmf-001-GPON_ACTIVATE_TARGET_ONT
Content-Type: application/json

{
  "operatorCode": "WIK",
  "commandType": "GPON_ACTIVATE_ONT",
  "subscriptionId": "SUB-700045",
  "homepassId": "HP-GPON-0440",
  "equipmentInstanceId": "inst-gpon-001",
  "originRefType": "TECHNOLOGY_MIGRATION_FULFILLMENT",
  "originRefId": "tmf-001",
  "payload": {
    "serialNumber": "ONT123456",
    "targetPackageRef": "WIK-FIBER-50M"
  }
}
```

9. Step 7 - Call FUL-09 When Equipment Swap Fulfillment Is Needed

If the equipment plan row says `CALL_FUL_09`, call:

```
POST /api/equipment-swap-fulfillments
Idempotency-Key: tmf-001-esf
Content-Type: application/json

{
  "operatorCode": "WIK",
  "swapRequestId": "tmf-001",
  "woId": "wo-mig-001",
  "subscriptionId": "SUB-700045",
  "accountId": "ACC-900045",
  "swapType": "EQUIPMENT_UPGRADE",
  "technologyFamily": "GPON",
  "sourceInstanceId": "inst-hfc-001",
  "targetInstanceId": "inst-gpon-001",
  "homepassId": "HP-GPON-0440"
}
```

Store returned id in `technology_migration_fulfillment.equipment_swap_fulfillment_id`.

Do not duplicate FUL-09 provisioning logic in FUL-10.

10. Step 8 - Reconcile

After provisioning commands:

1. Read target network state through PROV-INT read adapter.
2. Confirm target serial/MAC/ONT is active.
3. Confirm target service profiles match the target package.
4. Confirm source technology is suspended/teardown state according to cutover policy.
5. Confirm OSR-INSTANCE state is consistent or that OSR-RMA/FUL-09 has accepted responsibility for recovery.
6. Mark fulfillment `COMPLETED` only when reconciliation passes or an approved force-complete exists.

If reconciliation fails, create `technology_migration_exception`.

11. Step 9 - Callback SUB-WF

On success:

```
POST /internal/subscription-migrations/op-mig-001/fulfillment-outcome
Idempotency-Key: tmf-001-completed
Content-Type: application/json

{
  "migrationFulfillmentId": "tmf-001",
  "status": "COMPLETED",
  "sourceTechnology": "HFC",
  "targetTechnology": "GPON",
  "targetHomepassId": "HP-GPON-0440",
  "completedAt": "2026-08-04T11:30:00Z"
}
```

On failure:

```
{
  "migrationFulfillmentId": "tmf-001",
  "status": "FAILED",
  "failureCode": "TARGET_RECONCILIATION_FAILED",
  "exceptionId": "tmex-001"
}
```

SUB-WF decides whether to keep the operation pending, cancel it, or expose a repair path. FUL-10 does not update subscription master state.

12. Step 10 - Exception And Approval Flow

For force-complete, forced rollback, risky serial correction, or non-standard cutover:

1. Create `technology_migration_exception`.
2. Run exception routing rules.
3. If approval required, call EM-CFG-04.
4. Store `approval_request_id`.
5. Continue only after approval event/callback.

Sample rule:

```
package rules.fulfillment.technology_migration.exception

rule "Force complete after target verified requires approval"
when
  $f : TechnologyMigrationExceptionFact(exceptionType == "FORCE_COMPLETE",
                                        targetActivationVerified == true)
  $d : TechnologyMigrationExceptionDecision()
then
  $d.requireApproval("TECH_MIGRATION_FORCE_COMPLETE");
  $d.assignTeam("NOC_ACCESS");
  $d.setSeverity("HIGH");
end
```

Approval call:

```
POST /api/approval-requests
Content-Type: application/json

{
  "operatorCode": "WIK",
  "approvalCode": "TECH_MIGRATION_FORCE_COMPLETE",
  "originRefType": "TECHNOLOGY_MIGRATION_FULFILLMENT",
  "originRefId": "tmf-001",
  "requestedByUserId": "usr-noc-lead",
  "payload": {
    "exceptionId": "tmex-001",
    "targetActivationVerified": true,
    "sourceTeardownVerified": false
  }
}
```

Camunda is used only if EM-CFG-04 policy mode is `BPMN`.

13. Step 11 - Events And Outbox

Write all events to the transactional outbox in the same database transaction as the state change:

- `TechnologyMigrationFulfillmentRequested`
- `TechnologyMigrationValidated`
- `TechnologyMigrationWorkOrderCreated`
- `TechnologyMigrationCutoverStarted`
- `TechnologyMigrationCompleted`
- `TechnologyMigrationFailed`
- `TechnologyMigrationExceptionOpened`

Include:

- `operatorCode`
- `migrationFulfillmentId`
- `subscriptionOperationId`
- `subscriptionId`
- `correlationId`
- `causationId`

14. Step 12 - Tests

Implement tests in this order:

1. Start request creates fulfillment and is idempotent.
2. Same operation with different hash returns `409`.
3. Second active migration for same subscription returns `409`.
4. Target HomePass/package incompatibility creates failed validation.
5. HFC to GPON creates equipment plan with source recovery and target install.
6. WO creation sends correct `TECHNOLOGY_MIGRATION` context.
7. WO finalization with target serial enables cutover.
8. Cutover sends PROV-INT commands with deterministic idempotency keys.
9. Reconciliation pass completes and calls SUB-WF.
10. Reconciliation fail creates exception and does not callback success.
11. Force-complete calls EM-CFG-04 when configured.

15. Operational Notes

- Expose metrics by handler: validation failures, WO age, cutover duration, provisioning failures, reconciliation failures, approval wait time.
- Keep provisioning command request/response snapshots small and scrub secrets.
- The same worker application may implement all FUL-10 logical handlers.
- Do not let the frontend call provisioning adapters directly. Backoffice and support screens call FUL-10 APIs.
- Do not call OSR-02 procurement from this flow.